

# JUYEONG LEE

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## SUMMARY

AI engineer and quantitative researcher. At Qraft built multi-factor alpha, hedging and multi-asset allocation models for institutional investors; now at EY leading the ontology and multi-agent system behind a Korean athleisure brand's supply-chain decisions. Palantir Foundry & AIP certified; first author on DiffGerber and co-author on three further papers accepted at ICAIF 2026.

## EXPERIENCE

### EY Consulting (Ernst & Young)

Seoul, South Korea

*AI Consultant (Forward Deployed Engineer), AI & Data, Technology Consulting*

Jan 2026 – Present

*Andar (Korean athleisure brand): SCM decision system for forecasting, inventory and replenishment on Databricks and Neo4j*

- **SCM ontology & multi-agent system** — Leading the design and build: a Neo4j ontology of SKUs, seasons, promotions, reorder grades and weekly/monthly DIO, and a supervisor agent that routes planners' questions to four specialist agents for SCM monitoring, cost breakdown, raw and subsidiary material demand forecasting, and product demand forecasting; projected at full rollout to cut operating cost 15–20% and raise OTIF (on-time in-full) from 64% to 85%.
- **AI agents in use** — Led delivery of the three Genie agents (demand forecasting, days-of-inventory (DIO) and SCM agents) and the supervisor, used daily by planners; a 30-question golden set gates each release (accuracy up from 50% to 90%).
- **Demand forecasting model** — Led and built the product-level demand forecasting model with separate in-season and off-season versions; 83% forecast accuracy.
- **Samsung C&T Fashion PoC** — Led the PoC: built a merchandising ontology in Neo4j and a GraphRAG agent that answers multi-step merchandising questions and shows the supporting graph path; demoed to the client.

### Qraft Technologies (AI Hedge Fund)

Seoul, South Korea

*Researcher (AI / Quant), Investment Product Management*

Nov 2021 – May 2025

*Multi Asset Team*

Jul 2024 – May 2025

- **Client regime signals** — Built ML models that flag risk-on/off regimes from market, macro and stock-level data and delivered client-specific equity/cash allocation signals to Samsung Securities, KB Securities, Hana Life and Simplify Asset Management (US).
- **QIS strategy clustering** — Grouped quantitative strategies with PMFG graph-based hierarchical clustering to remove overlapping exposures and recombine them as regimes shift; validated in proofs of concept with global financial institutions.

*Long-Term Equity Team*

Nov 2021 – Jun 2024

- **SoftBank US equity long/short alpha pipeline** — Built the fund's statistical multi-factor algorithm by reducing 360 candidate strategies on financial-ratio, profitability and value factors to an orthogonal set: VIF, multiple regression and DBHT clustering removed collinear and overlapping factors before t-statistic and A/B screening; the fund delivered excess return over SPY (figures confidential).
- **Factor and risk model replication** — Reimplemented Fama-French (97% match) and the Barra risk model from scratch.
- **FSITC (Taiwanese asset manager) hedging framework** — Led the hedging-signal framework for its \$200M Japan fund: an ML, GNN and LSTM-autoencoder ensemble over equity, option-implied and market volatility; in a 2005–2024 TOPIX backtest it cut maximum drawdown from -62% to -33% and volatility from 21% to 11%.
- **Multi-agent risk reporting** — Automated FSITC's weekly risk report on top of the hedging signals with three LLM agents on Qraft's in-house harness: signal interpretation, market commentary and a risk manager that combines both into the risk assessment; also scored real-time news with chain-of-thought and few-shot prompting for a long/short strategy.
- **Trading simulator** — Built the backtesting simulator that models liquidity, slippage and market impact, raising agreement with realized P&L to 93%; used to backtest the SoftBank fund's strategies.

### UNIST Data Analytics Lab | NAVER Maps Joint Research Project

Ulsan, South Korea

*AI Researcher (MSc)*

Sep 2019 – Sep 2021

- **Traffic congestion prediction** — Led the joint research with NAVER Maps: built a time-series pattern-matching model combined with Graph Neural Networks (GCN/GAT) that predicts congestion after traffic incidents at 82% accuracy, 15% above the benchmark, and used LIME to explain the drivers; published as first author in Transportmetrica B.

## RESEARCH & PUBLICATIONS

- **ICAIF 2026** (ACM International Conference on AI in Finance; accepted), first author: DiffGerber — decision-focused learning of the Gerber covariance threshold from realized portfolio risk [\[paper\]](#) [\[slides\]](#)
- **Co-authored papers** (ICAIF 2026, accepted) — third author on AlphaLeak (“blind” LLM benchmark audit) [\[paper\]](#) [\[slides\]](#) and CallRank (language-model embeddings of earnings-call Q&A for sector rotation) [\[paper\]](#) [\[slides\]](#); co-author on Metropolitan Housing Signals for Treasury Duration Risk Management; SlipClock (crypto-futures execution-cost benchmark) under review.
- **Journals** (first author): *Quantifying incident impacts and identifying influential features in urban traffic networks*, Transportmetrica B, 2022 [\[link\]](#); *Prediction of problematic smartphone use: A machine learning approach*, IJERPH, 2021 [\[link\]](#)
- **Patents** (granted in Korea and Japan): *Predicting road congestion propagation using pattern matching*, lead inventor, KR 10-2608343 and JP 7453296 [\[link\]](#); *Predicting propagation time delay lag of road congestion using transfer entropy*, co-inventor, KR 10-2604575 and JP 7453297 [\[link\]](#)

## EDUCATION

**Ulsan National Institute of Science and Technology (UNIST)** — *MSc, Industrial Engineering*

Sep 2019 – Sep 2021

**Handong Global University** — *BA, Management and Economics (double major)*

Mar 2014 – Feb 2019

## SKILLS & CERTIFICATIONS

**Data & ontology:** Neo4j / Cypher, Databricks (Unity Catalog, Genie, Agent Bricks, Lakeview, Jobs), SQL / Spark SQL, Airflow, GCP

**AI / ML:** Python, PyTorch, TensorFlow, scikit-learn, pandas, LangChain, LangGraph, LLM agents, GNNs, time-series modelling

**Certifications & languages:** Palantir Foundry & AIP (Aug 2025); Korean (native); English (fluent; parliamentary interpreter, 2019)